

RP2350 Split Keyboard Firmware

A modular, C++ based firmware for an RP2350-powered split keyboard. This project integrates multiple hardware subsystems into a unified, non-blocking architecture.

Features

- **ADC Joystick Interface:** Configurable, math-driven transformation pipeline with an integrated Exponential Moving Average (EMA) filter for noise reduction.
- **LED Patterns:** (WIP) Dynamic visual feedback utilizing Programmable I/O (PIO).
- **Intra-communication:** (WIP) High-speed data transfer between split keyboard halves.
- **USB HID:** (WIP) Standardized host-computer interfacing.

Hardware Stack

- Microcontroller: Raspberry Pi Pico 2 / RP2350 (QFN-80 Package)
- C/C++ Pico SDK

Folder Structure

- `src/`: Contains the main application logic and entry point.
- `src/subsystems/`: Contains actively developed, project-specific hardware and logical modules (e.g., joystick, leds, comm).
- `config/`: Global headers and hardware pinout definitions.
- `docs/`: Project tracking, change logs, and architectural planning.
- `tests/`: Unit testing for independent modules.

Getting Started

1. Install the Pico C/C++ SDK.
2. Clone this repository.
3. Use CMake to generate the build files and compile the `.uf2` binary.